



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

INFORMATION RETRIEVAL [IT-4409]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the impact of Boolean retrieval on search engine efficiency compared to ranked retrieval models. (5)
 - A)
 - B) Interpret the effectiveness of different term normalization techniques (stemming vs. lemmatization) in index construction. (5)
- 2) Apply the impact of Heap's Law on vocabulary growth in large-scale information retrieval systems. How does it influence dictionary compression techniques? (5)
 - A)
 - B) Apply the cosine similarity measure and comment about influences document ranking in VSM.
 - Doc1: "Machine learning is powerful." (5)
 - Doc2: "Deep learning improves AI."
 - Doc3: "AI and machine learning are evolving."
- 3) Apply the precision-recall tradeoff concept to evaluate the effectiveness of a search system given the following dataset:
 - A)
 - Retrieved relevant documents: 20 (5)
 - Retrieved non-relevant documents: 30Total relevant documents in the corpus: 50
 - B) Compare relevance feedback and pseudo-relevance feedback. (5)
- 4) (5)
 - A)

For the query "*AI in Radiology*", consider the following term presence in documents:

Document	AI	Radiology
D1	1	0
D2	1	1
D3	0	1

1. Calculate RSV scores assuming:

$$\circ \quad df(AI) = 2, df(Radiology) = 2, N = 3.$$

Critically analyze why a document containing both terms (D2) should rank higher than others in a clinical context.

B) Discuss the search user experience in the case of a social media website. (5)

5) A search engine company wants to optimize its web crawler to improve efficiency and reduce redundant crawling. Given the following scenario:

- A)
- The company has a database of 1 billion URLs, but 20% are duplicate or near-duplicate.
 - The crawler operates 24/7 but experiences frequent slowdowns due to large web pages and dynamic content.
 - Some websites block the crawler due to excessive requests, leading to incomplete data collection.
- (5)

Interpret a crawling strategy that improves efficiency while maintaining comprehensive web coverage.

B) Discuss methods to handle duplicate content and reduce unnecessary crawling. (5)

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